

2-External SUGRA Bases

Colored curves in quiver show attachment points. Matching color in external label: **ext1** **ext2**

Total: 101 bases (0 uni, 101 non-uni).

T	Count
2	1
3	9
4	14
5	18
6	12
7	9
8	8
9	9
10	7
11	6
12	3
13	1
14	4

Non-unimodular ($|\det| > 1$) — 101 bases

$$\mathfrak{e}_6 + \hat{1} \rightarrow (-1)$$

$$T = 3 \quad [0] \quad 1^1 1^1 + \mathfrak{e}_6 + \hat{1} \rightarrow 1 \quad T = 3, T_{\max} = 13, \Delta = -291, \det = -6, (n_+, n_-, n_0) = (1, 3, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.67 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -5 & 0 & -3 \\ 1 & 0 & -1 & -1 \\ 2 & -3 & -1 & -3 \end{pmatrix}_{b_0}$$

$$T = 4 \quad [1] \quad 1^1 21^1 + \mathfrak{e}_6 + \hat{1} \rightarrow 1 \quad T = 4, T_{\max} = 13, \Delta = -262, \det = 5, (n_+, n_-, n_0) = (1, 4, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.80 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -4 & 0 & -2 \\ 1 & 0 & -1 & -1 \\ 2 & -2 & -1 & -2 \end{pmatrix}_{b_0}$$

$$T = 5 \quad [2] \quad 1^1 221^1 + \mathfrak{e}_6 + \hat{1} \rightarrow 1 \quad T = 5, T_{\max} = 13, \Delta = -233, \det = -4, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -3 & 0 & -1 \\ 1 & 0 & -1 & -1 \\ 2 & -1 & -1 & -1 \end{pmatrix}_{b_0}$$

$$\mathfrak{f}_4 + \mathfrak{f}_4 + \hat{1} \rightarrow (-1)$$

$$T = 7 \quad [0] \quad 1^1 321^1 + \mathfrak{f}_4 + \mathfrak{f}_4 + \hat{1} \rightarrow 1 \quad T = 7, T_{\max} = 14, \Delta = -210, \det = -8, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 17.4$$

$$T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & -3 & 0 & 1 & -1 \\ 1 & 0 & -5 & 0 & -3 \\ 1 & 1 & 0 & 2 & 2 \\ 2 & -1 & -3 & 2 & -1 \end{pmatrix}_{b_0}$$

$$\mathfrak{f}_4 + \hat{1} \rightarrow (-1)$$

$$T = 3 \quad [0] \quad 1^1 1^1 + \mathfrak{f}_4 + \hat{1} \rightarrow 1 \quad T = 3, T_{\max} = 12, \Delta = -265, \det = -5, (n_+, n_-, n_0) = (1, 3, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -4 & 0 & -2 \\ 1 & 0 & -1 & -1 \\ 2 & -2 & -1 & -2 \end{pmatrix}_{b_0}$$

$$T = 4 \quad [1] \quad 1^1 21^1 + \mathfrak{f}_4 + \hat{1} \rightarrow 1 \quad T = 4, T_{\max} = 12, \Delta = -236, \det = 4, (n_+, n_-, n_0) = (1, 4, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -3 & 0 & -1 \\ 1 & 0 & -1 & -1 \\ 2 & -1 & -1 & -1 \end{pmatrix}_{b_0}$$

$$\hat{1} \rightarrow (-1)$$

$$T = 2 \quad [0] \quad 1^{11}1^1 + \hat{1} \rightarrow 1 \quad T = 2, T_{\max} = 10, \Delta = -242, \det = 4, (n_+, n_-, n_0) = (1, 2, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 0 \\ 2 & 0 & 0 \end{pmatrix}_{b_0}$$

$$T = 3 \quad [2] \quad 1^{12}1^1 + \hat{1} \rightarrow 1 \quad T = 3, T_{\max} = 10, \Delta = -213, \det = -4, (n_+, n_-, n_0) = (1, 3, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$$

$$[1] \quad 21^{12} + \hat{1} \rightarrow 1 \quad T = 3, T_{\max} = 10, \Delta = -213, \det = -4, (n_+, n_-, n_0) = (1, 3, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$$

$$T = 4 \quad [4] \quad 1^{122}1^1 + \hat{1} \rightarrow 1 \quad T = 4, T_{\max} = 10, \Delta = -184, \det = 4, (n_+, n_-, n_0) = (1, 4, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[3] \quad 2^{12}2 + \hat{1} \rightarrow 1 \quad T = 4, T_{\max} = 10, \Delta = -184, \det = 4, (n_+, n_-, n_0) = (1, 4, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$T = 5 \quad [6] \quad 1^{1222}1^1 + \hat{1} \rightarrow 1 \quad T = 5, T_{\max} = 10, \Delta = -155, \det = -4, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 3 \\ 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[5] \quad 2^{22}21^1 + \hat{1} \rightarrow 1 \quad T = 5, T_{\max} = 10, \Delta = -155, \det = -4, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 3 \\ 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$T = 6 \quad [9] \quad 2^{31}21^1 + \hat{1} \rightarrow 1 \quad T = 6, T_{\max} = 11, \Delta = -145, \det = 4, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 3 \\ 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[8] \quad 1^{12222}1^1 + \hat{1} \rightarrow 1 \quad T = 6, T_{\max} = 10, \Delta = -126, \det = 4, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[7] \quad 2^{22}221^1 + \hat{1} \rightarrow 1 \quad T = 6, T_{\max} = 10, \Delta = -126, \det = 4, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 4 \end{pmatrix}_{b_0}$$

$$T = 7 \quad [12] \quad 2^{23}1321^1 + \hat{1} \rightarrow 1 \quad T = 7, T_{\max} = 11, \Delta = -117, \det = -4, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[11] \quad 1^{122222}1^1 + \hat{1} \rightarrow 1 \quad T = 7, T_{\max} = 10, \Delta = -97, \det = -4, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 5 \\ 2 & 5 & 5 \end{pmatrix}_{b_0}$$

$$[10] \quad 2^{22}2221^1 + \hat{1} \rightarrow 1 \quad T = 7, T_{\max} = 10, \Delta = -97, \det = -4, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 5 \\ 2 & 5 & 5 \end{pmatrix}_{b_0}$$

$$T = 8 \quad [13] \quad 2222221^1 + \hat{1} \rightarrow 1 \quad T = 8, T_{\max} = 10, \Delta = -68, \det = 4, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 6 & 6 \\ 2 & 6 & 6 \end{pmatrix}_{b_0}$$

$$[14] \quad 1^1 2222221^1 + \hat{1} \rightarrow 1 \quad T = 8, T_{\max} = 10, \Delta = -68, \det = 4, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 6 & 6 \\ 2 & 6 & 6 \end{pmatrix}_{b_0}$$

$$[15] \quad 2313221^1 + \hat{1} \rightarrow 1 \quad T = 8, T_{\max} = 11, \Delta = -88, \det = 4, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 5 \\ 2 & 5 & 5 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [16] \quad 22222221^1 + \hat{1} \rightarrow 1 \quad T = 9, T_{\max} = 10, \Delta = -39, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 7 & 7 \\ 2 & 7 & 7 \end{pmatrix}_{b_0}$$

$$[17] \quad 1^1 22222221^1 + \hat{1} \rightarrow 1 \quad T = 9, T_{\max} = 10, \Delta = -39, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 7 & 7 \\ 2 & 7 & 7 \end{pmatrix}_{b_0}$$

$$[18] \quad 31^1 5^{1231} 13 + \hat{1} \rightarrow 1 \quad T = 9, T_{\max} = 13, \Delta = -116, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[19] \quad 2^1 3^{123141} 2 + \hat{1} \rightarrow 1 \quad T = 9, T_{\max} = 12, \Delta = -87, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 5 \\ 2 & 5 & 5 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [20] \quad 222222221^1 + \hat{1} \rightarrow 1 \quad T = 10, T_{\max} = 10, \Delta = -10, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 8 \\ 2 & 8 & 8 \end{pmatrix}_{b_0}$$

$$[21] \quad 1^1 222222221^1 + \hat{1} \rightarrow 1 \quad T = 10, T_{\max} = 10, \Delta = -10, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 8 \\ 2 & 8 & 8 \end{pmatrix}_{b_0}$$

$$[22] \quad 31^1 5^{12231} 13 + \hat{1} \rightarrow 1 \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 5 \\ 2 & 5 & 5 \end{pmatrix}_{b_0}$$

$$[23] \quad 2^1 3^{1223141} 2 + \hat{1} \rightarrow 1 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 6 & 6 \\ 2 & 6 & 6 \end{pmatrix}_{b_0}$$

$$\mathfrak{so}_{16} \rightarrow \mathfrak{su}_8$$

$$T = 3 \quad [0] \quad 212^2 + \mathfrak{so}_{16} \rightarrow \mathfrak{su}_8 \quad T = 3, T_{\max} = 11, \Delta = -241, \det = -4, (n_+, n_-, n_0) = (1, 3, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 0 \\ 2 & 0 & 0 \end{pmatrix}_{b_0}$$

$$[1] \ 12^{\overline{2}1} + \textcolor{red}{s}\mathbf{o}_{16} \rightarrow \textcolor{red}{s}\mathbf{u}_8 \quad T = 3, \ T_{\max} = 11, \ \Delta = -241, \ \det = -4, \ (n_+, n_-, n_0) = (1, 3, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 0 \\ 2 & 0 & 0 \end{pmatrix}_{b_0}$$

$$\mathfrak{so}_8 + \mathfrak{e}_7 + \hat{1} \rightarrow (-1)$$

$$T = 8 \quad [0] \ 1^{\overline{1}2}321^{\overline{1}} + \textcolor{red}{s}\mathbf{o}_8 + \textcolor{blue}{e}_7 + \hat{1} \rightarrow 1 \quad T = 8, \ T_{\max} = 16, \ \Delta = -240, \ \det = 9, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 18.0$$

$$T_{\min} = 16, \ T_{\max} = 16, \ T \in [16, 16]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & -4 & 0 & 0 & -2 \\ 1 & 0 & -5 & 1 & -3 \\ 1 & 0 & 1 & 2 & 2 \\ 2 & -2 & -3 & 2 & -2 \end{pmatrix}_{b_0}$$

$$\mathfrak{so}_8 + \hat{1} \rightarrow (-1)$$

$$T = 3 \quad [0] \ 1^{\overline{1}1}1^{\overline{1}} + \textcolor{red}{s}\mathbf{o}_8 + \hat{1} \rightarrow 1 \quad T = 3, \ T_{\max} = 11, \ \Delta = -241, \ \det = -4, \ (n_+, n_-, n_0) = (1, 3, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -3 & 0 & -1 \\ 1 & 0 & -1 & -1 \\ 2 & -1 & -1 & -1 \end{pmatrix}_{b_0}$$

$$\mathfrak{so}_8 + \mathfrak{so}_8 + \hat{1} \rightarrow (-1)$$

$$T = 4 \quad [0] \ 1^{\overline{1}1}1^{\overline{1}} + \textcolor{red}{s}\mathbf{o}_8 + \textcolor{blue}{s}\mathbf{o}_8 + \hat{1} \rightarrow 1 \quad T = 4, \ T_{\max} = 12, \ \Delta = -240, \ \det = 16, \ (n_+, n_-, n_0) = (1, 4, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & -3 & 1 & 0 & -1 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 0 & 0 & -1 & -1 \\ 2 & -1 & -1 & -1 & -2 \end{pmatrix}_{b_0}$$

$$T = 5 \quad [1] \ 1^{\overline{1}1}21^{\overline{1}} + \textcolor{red}{s}\mathbf{o}_8 + \textcolor{blue}{s}\mathbf{o}_8 + \hat{1} \rightarrow 1 \quad T = 5, \ T_{\max} = 12, \ \Delta = -211, \ \det = -8, \ (n_+, n_-, n_0) = (1, 5, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & -2 & 2 & 0 & 0 \\ 1 & 2 & -2 & 0 & 0 \\ 1 & 0 & 0 & -1 & -1 \\ 2 & 0 & 0 & -1 & -1 \end{pmatrix}_{b_0}$$

$$\mathfrak{su}_2 + \mathfrak{e}_6 + \hat{1} \rightarrow (-1)$$

$$T = 4 \quad [0] \ 1^{\overline{1}1}1^{\overline{1}} + \textcolor{red}{s}\mathbf{u}_2 + \textcolor{blue}{e}_6 + \hat{1} \rightarrow 1 \quad T = 4, \ T_{\max} = 13, \ \Delta = -261, \ \det = 12, \ (n_+, n_-, n_0) = (1, 4, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.67 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & -4 & 0 & -2 \\ 1 & 0 & -1 & -1 \\ 3 & -2 & -1 & -2 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{1}1}1^{\overline{1}} + \textcolor{red}{s}\mathbf{u}_2 + \textcolor{blue}{e}_6 + \hat{1} \rightarrow 1 \quad T = 4, \ T_{\max} = 13, \ \Delta = -261, \ \det = 17, \ (n_+, n_-, n_0) = (1, 4, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.71 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 1 & 2 & 3 \\ 1 & -6 & 0 & -4 \\ 2 & 0 & 1 & 1 \\ 3 & -4 & 1 & -2 \end{pmatrix}_{b_0}$$

$$T = 5 \quad [2] \ 1^{\overline{1}1}21^{\overline{1}} + \textcolor{red}{s}\mathbf{u}_2 + \textcolor{blue}{e}_6 + \hat{1} \rightarrow 1 \quad T = 5, \ T_{\max} = 13, \ \Delta = -232, \ \det = -4, \ (n_+, n_-, n_0) = (1, 5, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 2 \\ 1 & 2 & -4 & 0 & -2 \\ 1 & 0 & 0 & -1 & -1 \\ 2 & 2 & -2 & -1 & -2 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{1}1}21^{\overline{1}} + \textcolor{red}{s}\mathbf{u}_2 + \textcolor{blue}{e}_6 + \hat{1} \rightarrow 1 \quad T = 5, \ T_{\max} = 13, \ \Delta = -232, \ \det = -14, \ (n_+, n_-, n_0) = (1, 5, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.86 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 2 \\ 1 & 0 & -6 & 0 & -4 \\ 1 & 2 & 0 & 1 & 1 \\ 2 & 2 & -4 & 1 & -2 \end{pmatrix}_{b_0}$$

$$\mathfrak{su}_2 + \mathfrak{f}_4 + \hat{1} \rightarrow (-1)$$

$$T = 4 \quad [0] \ 1^{\overline{1}1}1^{\overline{1}} + \textcolor{red}{s}\mathbf{u}_2 + \textcolor{blue}{f}_4 + \hat{1} \rightarrow 1 \quad T = 4, \ T_{\max} = 12, \ \Delta = -235, \ \det = 10, \ (n_+, n_-, n_0) = (1, 4, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & -3 & 0 & -1 \\ 1 & 0 & -1 & -1 \\ 3 & -1 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\hat{1}1}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{f}_4 + \hat{1} \rightarrow 1 \quad T = 4, T_{\max} = 12, \Delta = -235, \det = 14, (n_+, n_-, n_0) = (1, 4, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 1 & 2 & 3 \\ 1 & -5 & 0 & -3 \\ 2 & 0 & 1 & 1 \\ 3 & -3 & 1 & -1 \end{pmatrix}_{b_0}$$

$$\textcolor{red}{su}_2 + \hat{1} \rightarrow (-1)$$

$$T = 3 \quad [0] \ 1^{\hat{1}1}1^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 1 \quad T = 3, T_{\max} = 10, \Delta = -212, \det = -2, (n_+, n_-, n_0) = (1, 3, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 1 & 3 \\ 1 & -1 & -1 \\ 3 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\hat{1}1}1^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 1 \quad T = 3, T_{\max} = 10, \Delta = -212, \det = -8, (n_+, n_-, n_0) = (1, 3, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 \\ 3 & 1 & 1 \\ 3 & 1 & 1 \end{pmatrix}_{b_0}$$

$$T = 4 \quad [2] \ 1^{\hat{1}1}21^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 1 \quad T = 4, T_{\max} = 10, \Delta = -183, \det = 7, (n_+, n_-, n_0) = (1, 4, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 2 & 2 \\ 1 & 0 & 2 & 2 \\ 2 & 2 & 1 & 1 \\ 2 & 2 & 1 & 1 \end{pmatrix}_{b_0}$$

$$T = 5 \quad [3] \ 1^{\hat{1}1}221^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 1 \quad T = 5, T_{\max} = 10, \Delta = -154, \det = -6, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 2 & 2 \\ 1 & 1 & 3 & 3 \\ 2 & 3 & 2 & 2 \\ 2 & 3 & 2 & 2 \end{pmatrix}_{b_0}$$

$$T = 6 \quad [4] \ 1^{\hat{1}1}2221^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 1 \quad T = 6, T_{\max} = 10, \Delta = -125, \det = 5, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 2 & 2 \\ 1 & 2 & 4 & 4 \\ 2 & 4 & 3 & 3 \\ 2 & 4 & 3 & 3 \end{pmatrix}_{b_0}$$

$$T = 7 \quad [5] \ 1^{\hat{1}1}22221^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 1 \quad T = 7, T_{\max} = 10, \Delta = -96, \det = -4, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 2 & 2 \\ 1 & 3 & 5 & 5 \\ 2 & 5 & 4 & 4 \\ 2 & 5 & 4 & 4 \end{pmatrix}_{b_0}$$

$$T = 8 \quad [6] \ 1^{\hat{1}1}222221^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 1 \quad T = 8, T_{\max} = 10, \Delta = -67, \det = 3, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 2 & 2 \\ 1 & 4 & 6 & 6 \\ 2 & 6 & 5 & 5 \\ 2 & 6 & 5 & 5 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [7] \ 1^{\hat{1}1}2222221^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 1 \quad T = 9, T_{\max} = 10, \Delta = -38, \det = -2, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 2 & 2 \\ 1 & 5 & 7 & 7 \\ 2 & 7 & 6 & 6 \\ 2 & 7 & 6 & 6 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{\hat{1}1}22313^121^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 1 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 2 & 2 & 2 \\ 2 & 10 & 8 & 8 \\ 2 & 8 & 5 & 5 \\ 2 & 8 & 5 & 5 \end{pmatrix}_{b_0}$$

$$[9] \ 2^2313221^{\hat{1}1} + \textcolor{red}{su}_2 + \hat{1} \rightarrow 1 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -12, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 2 & 2 & 2 \\ 2 & 4 & 6 & 6 \\ 2 & 6 & 5 & 5 \\ 2 & 6 & 5 & 5 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [10] \quad 31 \quad 1^{11} 2231 \quad 5 \quad 13^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 1 \quad T = 11, T_{\max} = 13, \Delta = -58, \det = -11, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 3 & 2 & 2 \\ 3 & 7 & 7 & 7 \\ 2 & 7 & 5 & 5 \\ 2 & 7 & 5 & 5 \end{pmatrix}_{b_0}$$

$$\textcolor{blue}{su}_2 + \hat{1} \rightarrow (-1) + \hat{1} \rightarrow (-1)$$

$$T = 4 \quad [0] \quad 1^{11} 1^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 1 + \hat{1} \rightarrow 1 \quad T = 4, T_{\max} = 11, \Delta = -210, \det = 2, (n_+, n_-, n_0) = (1, 4, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 1 & 2 & 3 \\ 1 & -1 & 0 & -1 \\ 2 & 0 & 1 & 1 \\ 3 & -1 & 1 & 0 \end{pmatrix}_{b_0}$$

$$\textcolor{blue}{su}_2 + \textcolor{blue}{so}_{16} \rightarrow \textcolor{blue}{su}_8$$

$$T = 4 \quad [0] \quad 12^2 1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{so}_{16} \rightarrow \textcolor{blue}{su}_8 \quad T = 4, T_{\max} = 11, \Delta = -211, \det = 8, (n_+, n_-, n_0) = (1, 4, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 2 & 2 \\ 1 & 0 & 2 & 2 \\ 2 & 2 & 0 & 0 \\ 2 & 2 & 0 & 0 \end{pmatrix}_{b_0}$$

$$\textcolor{blue}{su}_2 + \textcolor{blue}{so}_8 + \hat{1} \rightarrow (-1)$$

$$T = 4 \quad [0] \quad 1^{11} 1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{so}_8 + \hat{1} \rightarrow 1 \quad T = 4, T_{\max} = 11, \Delta = -211, \det = 8, (n_+, n_-, n_0) = (1, 4, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & -2 & 0 & 0 \\ 1 & 0 & -1 & -1 \\ 3 & 0 & -1 & 0 \end{pmatrix}_{b_0}$$

$$\textcolor{blue}{su}_2 + \textcolor{blue}{so}_8 + \textcolor{blue}{so}_8 + \hat{1} \rightarrow (-1)$$

$$T = 5 \quad [0] \quad 1^{11} 1^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{so}_8 + \textcolor{green}{so}_8 + \hat{1} \rightarrow 1 \quad T = 5, T_{\max} = 12, \Delta = -210, \det = -40, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 1 & 1 & 2 & 3 \\ 1 & -4 & 0 & 0 & -2 \\ 1 & 0 & -4 & 0 & -2 \\ 2 & 0 & 0 & 1 & 1 \\ 3 & -2 & -2 & 1 & -1 \end{pmatrix}_{b_0}$$

$$T = 6 \quad [1] \quad 1^{11} 21^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{so}_8 + \textcolor{green}{so}_8 + \hat{1} \rightarrow 1 \quad T = 6, T_{\max} = 12, \Delta = -181, \det = 16, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 0 & 0 & 2 & 2 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 2 & 0 & 0 & 1 & 1 \\ 2 & 2 & -2 & -2 & 1 & -1 \end{pmatrix}_{b_0}$$

$$\textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \hat{1} \rightarrow (-1)$$

$$T = 6 \quad [0] \quad 1^{11} 221^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \hat{1} \rightarrow 1 \quad T = 6, T_{\max} = 10, \Delta = -124, \det = 5, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & 0 & 3 & 3 \\ 1 & 0 & -2 & 0 & 0 \\ 2 & 3 & 0 & 2 & 2 \\ 2 & 3 & 0 & 2 & 2 \end{pmatrix}_{b_0}$$

$$\textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) + \hat{1} \rightarrow (-1)$$

$$T = 12 \quad [0] \quad 31 \quad 1^{11} 2231 \quad 5 \quad 13^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) + \hat{1} \rightarrow 1 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 8, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 3 \\ 3 & 5 & 7 & 7 \\ 3 & 7 & 6 & 6 \\ 3 & 7 & 6 & 6 \end{pmatrix}_{b_0}$$

$$\mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{e}_6 + \hat{1} \rightarrow (-1)$$

$$T = 5 \quad [0] \quad 1^{11}1^{11} + \textcolor{red}{\mathfrak{su}_2} + \textcolor{blue}{\mathfrak{su}_3} + \textcolor{green}{\mathfrak{e}_6} + \hat{1} \rightarrow 1 \quad T = 5, T_{\max} = 13, \Delta = -240, \det = -45, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 1 & 1 & 2 & 3 \\ 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & -6 & 0 & -4 \\ 2 & 0 & 0 & 1 & 1 \\ 3 & -1 & -4 & 1 & -2 \end{pmatrix}_{b_0}$$

$$T = 6 \quad [1] \quad 1^{11}21^{11} + \textcolor{red}{\mathfrak{su}_2} + \textcolor{blue}{\mathfrak{su}_3} + \textcolor{green}{\mathfrak{e}_6} + \hat{1} \rightarrow 1 \quad T = 6, T_{\max} = 13, \Delta = -211, \det = 18, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 0 & 0 & 2 & 2 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 1 & 2 & 0 & 0 & 1 & 1 \\ 2 & 2 & -1 & -4 & 1 & -2 \end{pmatrix}_{b_0}$$

$$\mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_3 + \hat{1} \rightarrow (-1)$$

$$T = 6 \quad [0] \quad 1^{11}1^{11} + \textcolor{red}{\mathfrak{su}_2} + \textcolor{blue}{\mathfrak{su}_3} + \textcolor{green}{\mathfrak{su}_3} + \textcolor{orange}{\mathfrak{su}_3} + \hat{1} \rightarrow 1 \quad T = 6, T_{\max} = 11, \Delta = -149, \det = 54, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 1 & 1 & 1 & 2 & 3 \\ 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 0 & 0 & 0 & 1 & 1 \\ 3 & -1 & -1 & -1 & 1 & 0 \end{pmatrix}_{b_0}$$

$$\mathfrak{su}_2 + (-3) \rightarrow (-2) + \mathfrak{e}_7 + \hat{1} \rightarrow (-1)$$

$$T = 7 \quad [0] \quad 1^{11}22^{11} + \textcolor{red}{\mathfrak{su}_2} + (-3) \rightarrow (-2) + \textcolor{green}{\mathfrak{e}_7} + \hat{1} \rightarrow 1 \quad T = 7, T_{\max} = 15, \Delta = -238, \det = -23, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 17.0$$

$$T_{\min} = 14.91 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 0 & 3 & 3 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 1 & 0 & 0 & -8 & 0 & -6 \\ 1 & 3 & 1 & 0 & 2 & 2 \\ 2 & 3 & 0 & -6 & 2 & -3 \end{pmatrix}_{b_0}$$

$$\mathfrak{su}_2 + (-3) \rightarrow (-2) + \mathfrak{e}'_7 + \hat{1} \rightarrow (-1)$$

$$T = 7 \quad [0] \quad 1^{11}22^{11} + \textcolor{red}{\mathfrak{su}_2} + (-3) \rightarrow (-2) + \textcolor{green}{\mathfrak{e}'_7} + \hat{1} \rightarrow 1 \quad T = 7, T_{\max} = 14, \Delta = -210, \det = -18, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 17.0$$

$$T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 0 & 3 & 3 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 1 & 0 & 0 & -7 & 0 & -5 \\ 1 & 3 & 1 & 0 & 2 & 2 \\ 2 & 3 & 0 & -5 & 2 & -2 \end{pmatrix}_{b_0}$$

$$\mathfrak{su}_2 \rightarrow (-3) + \hat{1} \rightarrow (-1)$$

$$T = 7 \quad [0] \quad 2^3 21^{11} + \textcolor{red}{\mathfrak{su}_2} \rightarrow (-3) + \hat{1} \rightarrow 1 \quad T = 7, T_{\max} = 11, \Delta = -117, \det = -5, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 2 & 2 \\ 1 & -2 & 0 & 0 \\ 2 & 0 & 3 & 3 \\ 2 & 0 & 3 & 3 \end{pmatrix}_{b_0}$$

$$T = 8 \quad [2] \quad 1^1 2313^1 21^{11} + \textcolor{red}{\mathfrak{su}_2} \rightarrow (-3) + \hat{1} \rightarrow 1 \quad T = 8, T_{\max} = 11, \Delta = -88, \det = 4, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 2 & 2 \\ 1 & 0 & 2 & 2 \\ 2 & 2 & 4 & 4 \\ 2 & 2 & 4 & 4 \end{pmatrix}_{b_0}$$

[1] $2^2 3^1 1^1 2^1 + \text{su}_2 \rightarrow (-3) + \hat{1} \rightarrow 1$ $T = 8, T_{\max} = 11, \Delta = -90, \det = 4, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 2 & 2 & 2 \\ 2 & 1 & 3 & 3 \\ 2 & 3 & 4 & 4 \\ 2 & 3 & 4 & 4 \end{pmatrix}_{b_0}$$

$T = 9$ [3] $1^1 2^2 3^1 1^1 2^1 + \text{su}_2 \rightarrow (-3) + \hat{1} \rightarrow 1$ $T = 9, T_{\max} = 11, \Delta = -59, \det = -3, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 2 & 2 \\ 1 & 0 & 2 & 2 \\ 2 & 2 & 5 & 5 \\ 2 & 2 & 5 & 5 \end{pmatrix}_{b_0}$$

$T = 10$ [4] $1^1 2^3 1^1 1^1 4^1 3^1 2^1 + \text{su}_2 \rightarrow (-3) + \hat{1} \rightarrow 1$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 3, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 2 & 2 \\ 1 & 0 & 2 & 2 \\ 2 & 2 & 5 & 5 \\ 2 & 2 & 5 & 5 \end{pmatrix}_{b_0}$$

[5] $3^1 1^1 5^1 1^1 3^1 + \text{su}_2 \rightarrow (-3) + \hat{1} \rightarrow 1$ $T = 10, T_{\max} = 13, \Delta = -88, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 3 \\ 3 & 5 & 5 \\ 3 & 5 & 5 \end{pmatrix}_{b_0}$$

[6] $2^1 1^1 2^3 1^1 1^1 4^1 2 + \text{su}_2 \rightarrow (-3) + \hat{1} \rightarrow 1$ $T = 10, T_{\max} = 12, \Delta = -60, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 2 & 2 \\ 2 & 2 & 4 & 4 \\ 2 & 4 & 5 & 5 \\ 2 & 4 & 5 & 5 \end{pmatrix}_{b_0}$$

$T = 11$ [7] $1^1 2^3 1^1 1^1 4^1 3^1 2^1 1^1 + \text{su}_2 \rightarrow (-3) + \hat{1} \rightarrow 1$ $T = 11, T_{\max} = 12, \Delta = -29, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 2 & 2 \\ 1 & 0 & 2 & 2 \\ 2 & 2 & 6 & 6 \\ 2 & 2 & 6 & 6 \end{pmatrix}_{b_0}$$

[8] $3^1 1^1 5^1 1^1 3^1 + \text{su}_2 \rightarrow (-3) + \hat{1} \rightarrow 1$ $T = 11, T_{\max} = 13, \Delta = -59, \det = -3, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 3 \\ 3 & 6 & 6 \\ 3 & 6 & 6 \end{pmatrix}_{b_0}$$

$T = 12$ [9] $2^1 1^1 2^3 1^1 1^1 4^1 1^1 2 + \text{su}_2 \rightarrow (-3) + \hat{1} \rightarrow 1$ $T = 12, T_{\max} = 13, \Delta = -30, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.0$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 2 & 2 \\ 2 & 3 & 5 & 5 \\ 2 & 5 & 6 & 6 \\ 2 & 5 & 6 & 6 \end{pmatrix}_{b_0}$$

$T = 13$ [0] $3^1 2^1 8^1 1^1 2^3 2^1 + \text{su}_2 \rightarrow (-3) + \hat{1} \rightarrow 1$ $T = 13, T_{\max} = 16, \Delta = -87, \det = -4, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 8.0$

$$T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$$

$$\begin{pmatrix} 1 & 3 & 3 \\ 3 & 5 & 5 \\ 3 & 5 & 5 \end{pmatrix}_{b_0}$$

$T = 14$ [1] $2^1 1^1 2^3 1^1 1^1 4^1 1^1 1^1 2 + \text{su}_2 \rightarrow (-3) + \hat{1} \rightarrow 1$ $T = 14, T_{\max} = 14, \Delta = 0, \det = 4, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 8.0$

$$T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 0 & 2 & 2 & 2 \\ 2 & 4 & 6 & 6 \\ 2 & 6 & 7 & 7 \\ 2 & 6 & 7 & 7 \end{pmatrix}_{b_0}$$

$$\mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3) + \hat{1} \rightarrow (-1)$$

$$T = 8 \quad [0] \quad 2 \overset{3^{11}1}{3} 21^1 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3) + \hat{1} \rightarrow 1 \quad T = 8, T_{\max} = 11, \Delta = -90, \det = 4, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & -2 & 0 & 0 & 0 \\ 1 & 0 & -2 & 0 & 0 \\ 2 & 0 & 0 & 3 & 3 \\ 2 & 0 & 0 & 3 & 3 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [1] \quad 1^1 23^1 13^1 21^1 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3) + \hat{1} \rightarrow 1 \quad T = 9, T_{\max} = 11, \Delta = -61, \det = -3, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 2 & 2 \\ 1 & 2 & 1 & 3 & 3 \\ 2 & 2 & 3 & 4 & 4 \\ 2 & 2 & 3 & 4 & 4 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [2] \quad 1^1 23^1 1413^1 21^1 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3) + \hat{1} \rightarrow 1 \quad T = 11, T_{\max} = 12, \Delta = -31, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 0 & 0 & 2 & 2 \\ 1 & 0 & -2 & 0 & 0 \\ 2 & 2 & 0 & 5 & 5 \\ 2 & 2 & 0 & 5 & 5 \end{pmatrix}_{b_0}$$

$$[3] \quad 3^1 1^1 5^1 13^1 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3) + \hat{1} \rightarrow 1 \quad T = 11, T_{\max} = 13, \Delta = -60, \det = -3, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 1 & 3 & 3 \\ 1 & -2 & 0 & 0 \\ 3 & 0 & 5 & 5 \\ 3 & 0 & 5 & 5 \end{pmatrix}_{b_0}$$

$$T = 14 \quad [0] \quad 1^1 23^1 14141413^1 21^1 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3) + \hat{1} \rightarrow 1 \quad T = 14, T_{\max} = 14, \Delta = 0, \det = 0, (n_+, n_-, n_0) = (1, 14, 1), c_{\text{ext}} = 9.0$$

$$[1] \quad 1^1 2321^1 8^1 123^1 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3) + \hat{1} \rightarrow 1 \quad T = 14, T_{\max} = 16, \Delta = -60, \det = 3, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$$

$$\begin{pmatrix} 1 & 1 & 3 & 3 \\ 1 & -2 & 0 & 0 \\ 3 & 0 & 5 & 5 \\ 3 & 0 & 5 & 5 \end{pmatrix}_{b_0}$$

$$\mathfrak{su}_3 + \mathfrak{e}_6 + \hat{1} \rightarrow (-1)$$

$$T = 4 \quad [0] \quad 1^{11} 1^1 + \mathfrak{su}_3 + \mathfrak{e}_6 + \hat{1} \rightarrow 1 \quad T = 4, T_{\max} = 13, \Delta = -270, \det = 18, (n_+, n_-, n_0) = (1, 4, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & -5 & 0 & -3 \\ 1 & 0 & 0 & -1 & -1 \\ 2 & 0 & -3 & -1 & -3 \end{pmatrix}_{b_0}$$

$$T = 5 \quad [1] \quad 1^{11} 21^1 + \mathfrak{su}_3 + \mathfrak{e}_6 + \hat{1} \rightarrow 1 \quad T = 5, T_{\max} = 13, \Delta = -241, \det = -9, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 \\ 3 & 0 & 0 & 0 \\ 1 & 0 & -1 & -1 \\ 3 & 0 & -1 & -1 \end{pmatrix}_{b_0}$$

$$\mathfrak{su}_3 + \hat{1} \rightarrow (-1)$$

$$T = 7 \quad [0] \quad 1^{11} 2321^1 + \mathfrak{su}_3 + \hat{1} \rightarrow 1 \quad T = 7, T_{\max} = 11, \Delta = -116, \det = -5, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 2 & 2 \\ 1 & -3 & 0 & -1 \\ 2 & 0 & 7 & 5 \\ 2 & -1 & 5 & 3 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [1] \quad 1^{\overline{1}} 2 3 1^{\overline{1}} 5 1 3 2 1^{\overline{1}} + \textcolor{red}{\mathfrak{su}_3} + \hat{1} \rightarrow 1 \quad T = 11, \quad T_{\max} = 13, \quad \Delta = -59, \quad \det = -3, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 9.0$$

$$T_{\min} = 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 2 & 2 \\ 1 & -3 & 0 & -1 \\ 2 & 0 & 9 & 7 \\ 2 & -1 & 7 & 5 \end{pmatrix}_{b_0}$$

$$T = 12 \quad [2] \quad 1^{\overline{1}} 2 3 1^{\overline{1}} 5 1 3 2 2 1^{\overline{1}} + \textcolor{red}{\mathfrak{su}_3} + \hat{1} \rightarrow 1 \quad T = 12, \quad T_{\max} = 13, \quad \Delta = -30, \quad \det = 2, \quad (n_+, n_-, n_0) = (1, 12, 0), \quad c_{\text{ext}} = 9.0$$

$$T_{\min} = 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 2 & 2 \\ 1 & -3 & 0 & -1 \\ 2 & 0 & 10 & 8 \\ 2 & -1 & 8 & 6 \end{pmatrix}_{b_0}$$

$$\mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_3 + \hat{1} \rightarrow (-1)$$

$$T = 5 \quad [0] \quad 1^{\overline{1}} 1^{\overline{1}} 1^{\overline{1}} + \textcolor{red}{\mathfrak{su}_3} + \textcolor{blue}{\mathfrak{su}_3} + \textcolor{green}{\mathfrak{su}_3} + \hat{1} \rightarrow 1 \quad T = 5, \quad T_{\max} = 11, \quad \Delta = -179, \quad \det = -27, \quad (n_+, n_-, n_0) = (1, 5, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 11, \quad T_{\max} = 11, \quad T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 1 & 0 & 0 \\ 1 & 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 1 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 2 & 0 & 0 & 0 & -1 & -1 \end{pmatrix}_{b_0}$$

$$(-3) \rightarrow (-2) + \mathfrak{e}_7 + \hat{1} \rightarrow (-1)$$

$$T = 5 \quad [0] \quad 1^{\overline{1}} 2^{\overline{1}} 1^{\overline{1}} + (-3) \rightarrow (-2) + \textcolor{blue}{\mathfrak{e}_7} + \hat{1} \rightarrow 1 \quad T = 5, \quad T_{\max} = 15, \quad \Delta = -297, \quad \det = -21, \quad (n_+, n_-, n_0) = (1, 5, 0), \quad c_{\text{ext}} = 16.0$$

$$T_{\min} = 14.90 \rightarrow 15, \quad T_{\max} = 15, \quad T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & -6 & 0 & -4 \\ 1 & 0 & 0 & -1 & -1 \\ 2 & 0 & -4 & -1 & -4 \end{pmatrix}_{b_0}$$

$$T = 6 \quad [1] \quad 1^{\overline{1}} 2^{\overline{1}} 2 1^{\overline{1}} + (-3) \rightarrow (-2) + \textcolor{blue}{\mathfrak{e}_7} + \hat{1} \rightarrow 1 \quad T = 6, \quad T_{\max} = 15, \quad \Delta = -268, \quad \det = 11, \quad (n_+, n_-, n_0) = (1, 6, 0), \quad c_{\text{ext}} = 16.0$$

$$T_{\min} = 14.91 \rightarrow 15, \quad T_{\max} = 15, \quad T \in [15, 15]$$

$$\begin{pmatrix} 10 & 1 & 0 \\ 1 & -1 & -1 \\ 0 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{\overline{1}} 2^{\overline{1}} 2 1^{\overline{1}} + (-3) \rightarrow (-2) + \textcolor{blue}{\mathfrak{e}_7} + \hat{1} \rightarrow 1 \quad T = 6, \quad T_{\max} = 15, \quad \Delta = -268, \quad \det = 18, \quad (n_+, n_-, n_0) = (1, 6, 0), \quad c_{\text{ext}} = 16.0$$

$$T_{\min} = 15, \quad T_{\max} = 15, \quad T \in [15, 15]$$

$$\begin{pmatrix} 1 & 1 & 3 & 3 \\ 1 & -8 & 0 & -6 \\ 3 & 0 & 6 & 4 \\ 3 & -6 & 4 & -2 \end{pmatrix}_{b_0}$$

$$(-3) \rightarrow (-2) + \mathfrak{e}'_7 + \hat{1} \rightarrow (-1)$$

$$T = 5 \quad [0] \quad 1^{\overline{1}} 2^{\overline{1}} 1^{\overline{1}} + (-3) \rightarrow (-2) + \textcolor{blue}{\mathfrak{e}'_7} + \hat{1} \rightarrow 1 \quad T = 5, \quad T_{\max} = 14, \quad \Delta = -269, \quad \det = -18, \quad (n_+, n_-, n_0) = (1, 5, 0), \quad c_{\text{ext}} = 16.0$$

$$T_{\min} = 14, \quad T_{\max} = 14, \quad T \in [14, 14]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & -5 & 0 & -3 \\ 1 & 0 & 0 & -1 & -1 \\ 2 & 0 & -3 & -1 & -3 \end{pmatrix}_{b_0}$$

$$T = 6 \quad [1] \quad 1^{\overline{1}} 2^{\overline{1}} 2 1^{\overline{1}} + (-3) \rightarrow (-2) + \textcolor{blue}{\mathfrak{e}'_7} + \hat{1} \rightarrow 1 \quad T = 6, \quad T_{\max} = 14, \quad \Delta = -240, \quad \det = 9, \quad (n_+, n_-, n_0) = (1, 6, 0), \quad c_{\text{ext}} = 16.0$$

$$T_{\min} = 14, \quad T_{\max} = 14, \quad T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 \\ 3 & 0 & 0 & 0 \\ 1 & 0 & -1 & -1 \\ 3 & 0 & -1 & -1 \end{pmatrix}_{b_0}$$

$$(-3) \rightarrow (-2) + \mathfrak{e}_8 + \hat{1} \rightarrow (-1)$$

$$T = 6 \quad [0] \quad 1^{\hat{1}} 2^{\hat{1}} 21^{\hat{1}} + (-3) \rightarrow (-2) + \mathfrak{e}_8 + \hat{1} \rightarrow 1 \quad T = 6, \quad T_{\max} = 19, \quad \Delta = -383, \quad \det = 30, \quad (n_+, n_-, n_0) = (1, 6, 0), \quad c_{\text{ext}} = 17.0$$

$$T_{\min} = 18.73 \rightarrow 19, \quad T_{\max} = 19, \quad T \in [19, 19]$$

$$\begin{pmatrix} 1 & 1 & 3 & 3 \\ 1 & -12 & 0 & -10 \\ 3 & 0 & 6 & 4 \\ 3 & -10 & 4 & -6 \end{pmatrix}_{b_0}$$

$$(-3) \rightarrow (-2) + \hat{1} \rightarrow (-1)$$

$$T = 14 \quad [0] \quad 2^{\hat{1}} 1^{\hat{1} 2321} 8^{\hat{1}} 12321^{\hat{1}} + (-3) \rightarrow (-2) + \hat{1} \rightarrow 1 \quad T = 14, \quad T_{\max} = 16, \quad \Delta = -58, \quad \det = 3, \quad (n_+, n_-, n_0) = (1, 14, 0), \quad c_{\text{ext}} = 9.0$$

$$T_{\min} = 16, \quad T_{\max} = 16, \quad T \in [16, 16]$$

$$\begin{pmatrix} 1 & 3 & 3 \\ 3 & 6 & 6 \\ 3 & 6 & 6 \end{pmatrix}_{b_0}$$

$$\mathfrak{su}_8 \rightarrow \mathfrak{so}_{16}$$

$$T = 5 \quad [1] \quad 14^{\hat{2} 1} + \mathfrak{su}_8 \rightarrow \mathfrak{so}_{16} \quad T = 5, \quad T_{\max} = 11, \quad \Delta = -183, \quad \det = -4, \quad (n_+, n_-, n_0) = (1, 5, 0), \quad c_{\text{ext}} = 1.5$$

$$T_{\min} = 11, \quad T_{\max} = 11, \quad T \in [11, 11]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[0] \quad 14^{\hat{2} 1} + \mathfrak{su}_8 \rightarrow \mathfrak{so}_{16} \quad T = 5, \quad T_{\max} = 11, \quad \Delta = -183, \quad \det = -4, \quad (n_+, n_-, n_0) = (1, 5, 0), \quad c_{\text{ext}} = 1.5$$

$$T_{\min} = 11, \quad T_{\max} = 11, \quad T \in [11, 11]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[2] \quad 14^{\hat{2} 122} + \mathfrak{su}_8 \rightarrow \mathfrak{so}_{16} \quad T = 5, \quad T_{\max} = 11, \quad \Delta = -183, \quad \det = -4, \quad (n_+, n_-, n_0) = (1, 5, 0), \quad c_{\text{ext}} = 1.5$$

$$T_{\min} = 11, \quad T_{\max} = 11, \quad T \in [11, 11]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[3] \quad 214^{\hat{2} 12} + \mathfrak{su}_8 \rightarrow \mathfrak{so}_{16} \quad T = 5, \quad T_{\max} = 11, \quad \Delta = -183, \quad \det = -4, \quad (n_+, n_-, n_0) = (1, 5, 0), \quad c_{\text{ext}} = 1.5$$

$$T_{\min} = 11, \quad T_{\max} = 11, \quad T \in [11, 11]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[4] \quad 22214^{\hat{2}} + \mathfrak{su}_8 \rightarrow \mathfrak{so}_{16} \quad T = 5, \quad T_{\max} = 11, \quad \Delta = -183, \quad \det = -4, \quad (n_+, n_-, n_0) = (1, 5, 0), \quad c_{\text{ext}} = 1.5$$

$$T_{\min} = 11, \quad T_{\max} = 11, \quad T \in [11, 11]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{pmatrix}_{b_0}$$